



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (CSE) (Credit System) Degree Examinations

Supplementary Examinations – August 2025

21CS701- BIG DATA ANALYTICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit– I** and **Unit – II** each and **One full** question from **Unit – III**.*

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	List and Explain the Data management waves. Explain with a neat diagram cycle of Big Data Management.	08	L*2	1	1
	b)	Explain the Big Data architecture in detail with a neat diagram.	06	L2	1	1
	c)	List and explain various sources of human generated and machine generated unstructured data.	06	L2	1	1
2.	a)	What are the various challenges in layer0 and layer1.	08	L1	1	1
	b)	Write a short note on role of relational databases in big data.	04	L1	1	1
	c)	List and explain the various tools used in the layer3 and layer 4.	08	L2	1	1
3.	a)	Compare and contrast structured and unstructured data. List and explain the sources of human generated and machine generated structured data.	10	L3	1	1
	b)	Explain the features of operational data sources.	04	L2	1	1
	c)	Compare and Contrast Traditional and advanced Analytics. List and explain three V's of big data.	06	L2	1	1
		Unit – II				
4.	a)	Describe the properties of nonrelational databases.	08	L2	2	1
	b)	Compare and contrast the features of NoSQL and NewSQL.	04	L2	3	1
	c)	List and explain the important features of HBase columnar databases.	08	L2	2	1
5.	a)	List and explain advantages and disadvantages of NoSQL.	08	L2	3	1
	b)	Compare and contrast MongoDB and CouchDB databases.	08	L2	2	1
	c)	Write a short note on key-value pair databases.	04	L2	2	1
6.	a)	Discuss the fundamental behaviors of MapReduce. Illustrate dataflow in MapReduce with a neat diagram.	08	L2	4	1
	b)	Describe the features of Neo4J graph database.	08	L2	3	1
	c)	Describe various features of PostgreSQL databases.	04	L2	3	1
		Unit – III				
7.	a)	With the help of suitable examples explain basic analytics and advanced analytics.	06	L2	5	1
	b)	Illustrate the architecture of Hadoop Distributed File System with the help of a neat diagram.	08	L2	5	1
	c)	Discuss the concept of monetizing analytics.	06	L2	5	1
8.	a)	List and explain any four key features of Sqoop.	04	L2	5	1
	b)	Illustrate the YARN architecture with a neat diagram. Explain the services provided by Yet Another Resource Negotiator.	08	L3	5	1
	c)	Explain the potential characteristics of data.	08	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
